

Dr. Tobias Marc Leva

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Profile

Neuroscientist with 5+ years of hands-on in-vivo research experience, combining deep expertise in electrophysiology, neuromodulation, and neural engineering with a strong drive to turn the science-fiction vision of biobots into reality, while collaboratively establishing Swarm Biotactics as a pioneering force in the field.

Experience

Max Delbrück Center for Molecular Medicine
AG Poulet - Neural Circuits and Behavior
Researcher

Berlin, Germany

May 2017 – Present

- **Demonstrated expertise in in-vivo systems-neuroscience research**, incl. acute Neuropixels recordings, software-based quantification of neuroanatomy and its reconstruction, quantitative investigation of behaviour and sensory perception during neuromodulation (optogenetics, pharmacogenetics, pharmacology), and Python-based data processing, analysis, and visualisation, uncovering novel principles of how the brain creates temperature perception.
- **Drove scientific product development** by designing, engineering, building, and iteratively optimizing a cutting-edge high-density extracellular electrophysiology setup with superior recording efficiency compared to standard rigs, enabling unprecedented investigation of how the thalamus encodes temperature by acquiring neural activity from 15000+ neurons in awake mice.
- **Identified, mapped, and quantified previously unknown neural pathways** by conducting neuroanatomical investigations in the mammalian brain, including brain-wide tracing from primary sensory cortices, immunohistochemistry, imaging of labeled brain tissue, and reconstruction/alignment of anatomical image data to a virtual brain template for enhanced visualization and analysis.
- **Enabled systematic investigation of neural circuits by applying hands-on microsurgical techniques**, including stereotactic implantation of head-holders for in vivo extracellular recordings in awake mice, craniotomy and durotomy, stereotactic microinjections, and implantation of RFID chips in the abdominal cavity of mice, serving as the methodological foundation for my scientific work.
- **Owned the entire scientific project lifecycle**, from formulating the research question through execution, analysis, and presentation/publishing, while actively collaborating with fellow project leads; the work culminated in multiple publications in high-impact neuroscience journals.
- **Developed open-source Python codes and pipelines and maintained GitHub repository** for data acquisition, preprocessing, signal analysis, and visualisation of neural data (>400 GB/day), enabling high-throughput workflows.
- **Demonstrated strong communication and presentation skills** of complex technological concepts and scientific results to diverse audiences, as evidenced by multiple conference presentations and teaching responsibilities.
- **Led multidisciplinary teams of scientists**, engineers, and technicians; **set milestones, tracked progress, mentored junior members**, and cultivated a results-driven lab culture resulting in co-authorships on my scientific publications.

Centre for Genomic Regulation (CRG)
AG Louis - Sensory Systems and Behavior
Research Assistant

Barcelona, Spain

May 2016 – Oct 2016

- Collaborative development of **locomotion model and its relationship to neural activity in Drosophila larvae**, incl. manipulations of neural activity via Optogenetics, high-speed video recordings for locomotion tracking, and extraction of movement features from videos using custom Matlab scripts
- **Verified expression of optogenetic actuators in motor neurons** by performing microdissection of larval brains and applying advanced imaging techniques (2-photon, confocal), **ensuring accurate targeting of genetically defined cell populations**.

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Blies Festival ([Link](#)) Head of Production

Ludwigshafen, Germany
2023 & 2024

- **Led the pre-production, production, and post-production** of a music and culture festival for **3000+ attendees**
- Ensured successful project by **executing strategic and logistic planning, team recruiting, budgeting, stakeholder communication, and fundraising**, securing over € 60k+ in funding
- Independently performed post-production, including attendee **demographics analysis, conducting business analyses of costs and revenues, and delivering a final stakeholder and sponsor presentation**, resulting in strategic insights and sustainable funding for future events.

'Badias' Restaurant in der Schirn Kunsthalle ([Link](#)) Co-founder and Manager

Frankfurt, Germany
Aug 2015 – Feb 2016

- Implementation of sustainable operations that led to a **decade of continued success**
- **Recruited a diverse team** of employees, **optimized operational workflows**, and **implemented strategic growth plans**
- Time and **team management**, budgeting

Education

Humboldt University

Berlin, Germany
22.02.2024

Ph.D. Neuroscience, *Magna cum laude* [GPA: 3.9 / 4.0]

Thesis: The Spatial Representation of Temperature across the Thalamus (MDC Berlin)

NeuroCure – Cluster of Excellence Research Fellowship: 10 awardees/year (top 1%), full tuition and stipend for 3 years.

Tutored +20 master's students on 'Data Analysis in Python' for 2 semesters, covering mathematical and statistical basics and hands-on programming in Python. Published first and co-authored papers in high-impact neuroscience journals.

Two invited presentations at international science conferences.

Goethe University

Frankfurt, Germany
04.05.2015

Diplom (M.Sc. equivalent) Biochemistry [GPA: 3.9 / 4.0]

Majors: Biophysical Chemistry, Neurobiology, Biochemistry

Thesis: Antenna complexes in microbial rhodopsins (Max Planck Institute for Biophysics)

Tutored +40 bachelor's students on 'Principles of Metabolism' and 'Kinetics and Electrostatics' for 2 semesters.

Certificates:

- TensorFlow Developer: Neural Networks, CNN, NLP, Time Series & Prediction
- Data Scientist: EDA, Machine learning, Model Evaluation, Performance Metrics
- English Proficiency: Score: 105 / 120

DeepLearning.ai (2024)
neueFische (2023)
TOEFL iBT (2015)

Personal Information & Skills

Personality: Motivated, Analytical, Team Player, Adaptability, Communicator, Friendly, Problem-Solver, Hands-on, Creative

Languages: German (native), English (excellent), French (basic)

Neuroscience Research: Neuroanatomy (Injections, imaging, quantification), Sensory Encoding, Neuromodulation (Optogenetics, pharmacogenetics, electrical, pharmacological), Behavioral testing and quantification, Microsurgery

Data Science: Python, Mathematical and Statistical Analysis (numpy, scipy), Neural Networks (TensorFlow), Machine Learning (Scikit-learn), CV (openCV, DeeplabCut), DataViz (matplotlib, seaborn), Data Management (Pandas, Neo, SQL), Project Management (Asana)

Neural Engineering: High-density Ephys. (Neuropixels), Device Development, Hardware- Software Interfacing & Device Synchronisation (National Instruments, Arduino),

Interests: Woodworking, Furniture Design, Sports (Tennis, Yoga, Golf), Cooking & Hosting, Humor & Laughter

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Publications & Conference Presentations

Tobias M Leva, Clarissa J. Whitmire, Ilaria Sauve, Phillip Bokinić, Charlene Memler, Bibiana M. Horn, Mikkel Vestergaard, Mario Carta, James F.A. Poulet, **The cellular representation of temperature across the somatosensory thalamus**, bioRxiv 2024.02.13.580167; <https://doi.org/10.1101/2024.02.13.580167>

Tobias M Leva, Clarissa J Whitmire, **Thermosensory thalamus: parallel processing across model organisms**, Front. Neurosci. Sec. Translational Neuroscience Volume 17 – 2023, 13 October 2023, <https://doi.org/10.3389/fnins.2023.1210949>

Phillip Bokinić, Clarissa J Whitmire, **Tobias M Leva**, James F A Poulet, **Brain-wide connectivity map of mouse thermosensory cortices**, Cerebral Cortex, Volume 33, Issue 8, 15 April 2023, Pages 4870–4885, <https://doi.org/10.1093/cercor/bhac386>

Conference Presentations

- GRC Thalamocortical Interactions (2022) **Distinct Cellular Encoding of Warm and Cool in the Thalamus**, Lucca, IT
- 35th Annual Barrels Meeting (2022) **Distinct Cellular Encoding of Warm and Cool in the Thalamus**, La Jolla, US